

PHAROS — Phase 3 Remaining Issues & Punch List

Generated: 2026-06-20 10:33 IST Branch: **main** System: **PHAROS v3 (Node.js/Express + PostgreSQL + RabbitMQ + React/Vite)**

1 — Module Completion Matrix

MODULE	BACKEND	ROUTER / SCOPE	FRONTEND PAGE	INTEGRATION
Report Builder	✓ Full (queryEngine + controller)	✓ <code>/api/reports/builder/*</code> enforceScope ✓	✓ ReportBuilder.jsx (1260 lines)	✓ All API calls match routes
Legacy Import	✓ Full (863 lines)	✓ <code>/api/legacy/*</code> enforceScope ✓ (fixed 2026-06-20)	✓ LegacyDataPage.jsx (437 lines)	✓ API calls match routes
Bulk Import (Validate/Confirm)	✓ Full (864 lines)	✓ <code>/api/import/*</code> enforceScope ✓ (fixed 2026-06-20)	✗ No frontend page	✗ API is headless UI
Warehouse ETL	✓ ETL pipeline + scheduler	✓ <code>/api/warehouse/status</code> only	✗ No UI (not critical)	✓ Scheduler runs on startup
Filter Presets	✓ CRUD controller	✓ <code>/api/filters/presets</code> enforceScope ✓ (fixed 2026-06-20)	⚠ Hook exists, no UI component	⚠ Hook wired to page
Level Contracts	✓ service + controller	✓ <code>/api/level-contracts</code> enforceScope ✓ (fixed 2026-06-20)	✓ LevelContractsPage.jsx (290 lines)	✓ Full round trip
New Form Fields (24 fields)	✓ In seed.js code (commit 142d1c7)	N/A	N/A (DynamicForm auto-renders)	✗ Seed run again yet
Daily Diary	✓ Fully done	✓ <code>/api/daily-diary/*</code>	✓ CompilationUI.jsx	✓

2 — Critical Blockers

BLOCKER Destructive migration will wipe report_jobs on next run

File: `backend/migrations/20260620300000_phase3_reports_import.js`

The `up()` function calls `knex.schema.dropTableIfExists('report_jobs')` — this deletes all pending/completed report jobs in production on every migration run. Must be rewritten to use `alterTable` or check for column existence before dropping.

Do NOT run this migration as-is.

BLOCKER Untracked migration file not committed

File: `backend/migrations/20260619000000_nullable_template_id_report_jobs.js`

This migration exists on disk but has never been committed to git. Any developer pulling the repo will not have it. Must be `git add` + committed before next deployment.

BLOCKER 24 new form fields not in database

Fields for CASE (C_29–C_35), ARREST (A_24–A_29), MISSING (MS_23–MS_28), UIDB (U_14–U_28) were added to `backend/seeds/seed.js` in commit `142d1c7` but the seed has not been re-run against the database. Forms will not show these fields until seed runs. In Docker setup, this happens at container start via `start.bat` — but requires seed to be idempotent (`onConflict('field_key').ignore()`).

3 — High Priority Items

HIGH Bulk Import has no frontend page

The `/api/import/*` module provides template download, batch validation (dry-run with error report), and batch confirmation. It is fully functional but completely invisible to users — no React page exists. This is a core Phase 3 feature (legacy CSV/XLSX import flow with validation step). Needs a new page at `/import` or wired into `LegacyDataPage.jsx` as a second tab.

HIGH Possible duplicate notification subscriptions

`notifyHandler.init()` is called in `app.js` and subscribes to `record.submitted`, `record.approved`, `record.sent_back`, `compilation.submitted`.
`notifications.service.js` `initSubscriptions()` is called in `index.js` and may subscribe to the same events.
In RabbitMQ mode, competing consumers on the same queue would split messages. In mock/in-memory mode, both handlers fire — doubling notifications. Needs audit and consolidation into one handler.

HIGH Filter presets hook not wired to any UI

`frontend/src/hooks/useFilterPresets.js` wraps the filter presets API but is not imported or used by any page component. The records list page and analytics page have no way for users to save/load filter presets. Needs integration into the records queue or analytics page filter panel.

4 — Medium Priority Items

MEDIUM Warehouse status monitoring UI absent

Only `GET /api/warehouse/status` exists. No dashboard widget or admin page shows last sync time, record counts, or sync health. Recommended: add a small status card to the HQ or Admin dashboard showing warehouse readiness and last sync timestamp.

MEDIUM report-builder sub-mount not dual-registered

The report builder router is mounted via `reports.router.js` → `router.use('/builder', reportBuilderRouter)`. This means it's accessible at `/api/reports/builder/*` and `/api/v1/reports/builder/*` (both work because `reports.router` itself is dual-mounted). Confirmed correct — no action needed, just documented for clarity.

MEDIUM import.router.js uses its own Multer config

The import router declares its own `multer.diskStorage` config with a `./temp-imports` directory, instead of using the shared `backend/src/middleware/upload.middleware.js`. Minor inconsistency; not a bug, but creates two upload paths. Consider consolidating.

MEDIUM JCP/SCP roles missing from enforceScope

`rbac.middleware.js enforceScope()` handles HC/SHO (`ps_id`), DISTRICT_OFFICER (`district_id`), ACP (`sub_div_id`), and HQ/SYSTEM (global). JCP and SCP roles are not handled — they fall through to empty `jurisdictionQuery` (same as HQ). If JCP/SCP users are ever added, they will see all records. Phase 3 plan includes JCP/SCP chain — add cases to `enforceScope` before onboarding those roles.

5 — Low Priority / Cleanup

LOW All console.* replaced with logger (done 2026-06-20)

59 console.log/warn/error calls across 12 files were replaced with `logger.info/warn/error/debug` from `utils/logger.js`. Files fixed: `app.js`, `index.js`, `auditHandler.js`, `notifyHandler.js`, `auth.service.js`, `auth.middleware.js`, `compilation.service.js`, `reports.controller.js`, `scheduler.js`, `import.controller.js`, `warehouse.db.js`, `warehouse/etl/backfill.js`, `warehouse/etl/dimensions.js`, `warehouse/etl/sync.js`, `utils/s3.js`.

LOW Old orphaned *.routes.js files still in repo

Several modules have both `*.router.js` (active, imported by `app.js`) and `*.routes.js` (orphaned, not imported). Examples: `compilation.routes.js`, `notifications.routes.js`. They should be deleted to avoid confusion — but verify no other file imports them first.

LOW Vaibhav's branch naming violated the brief

PR #16 came from `feature/daily-diary-compilation` instead of `feature/be-reports` / `feature/be-import` as specified. Vaibhav self-merged without owner review. Process enforcement needed: require PR approval from repo owner (`Ashmit72061`) before merge.

6 — Phase 3 Roadmap Items Not Yet Started

ITEM	FROM ROADMAP	EFFORT	STATUS
MinIO integration (file attachments)	Phase 3 infra	Medium	NOT STARTED
Offline PWA support	Phase 3	Large	NOT STARTED
Elasticsearch integration (full-text search)	Phase 3	Large	NOT STARTED
Performance testing (250 PS load)	Phase 3	Large	NOT STARTED
Analytics: PS comparison, beat-wise views	Phase 2/3	Medium	NOT STARTED
Hash-sealed audit (chain verification API)	Phase 2	Medium	HASH WRITTEN, NO VERIFY API
Keycloak / MFA integration	Phase 2	Large	STUB ONLY (falls back to JWT)
Scheduled reports UI (HQ/Admin cron)	Phase 3	Medium	BACKEND DONE, FRONTEND MISSING

7 — RabbitMQ / Event Bus Status

RabbitMQ workers are correctly initialized. Both `auditHandler.init()` and `notifyHandler.init()` are called in `app.js startServer()` after `eventBus.connect()`. The warehouse ETL does not use RabbitMQ — it uses `node-cron`, which is correct for a scheduled sync workload.

HANDLER / WORKER	INITIALIZED IN	SUBSCRIBES TO	STATUS
<code>auditHandler</code>	<code>app.js → startServer()</code>	<code>record.*</code> → audit-queue → writes <code>record_revisions</code>	✓ WIRED
<code>notifyHandler</code>	<code>app.js → startServer()</code>	<code>record.submitted</code> , <code>record.approved</code> , <code>record.sent_back</code> , <code>compilation.submitted</code> → SSE push	✓ WIRED
<code>notifications.service</code> <code>initSubscriptions</code>	<code>index.js → start()</code>	Unknown — may duplicate <code>notifyHandler</code> subscriptions	⚠ AUDIT NEEDED
Warehouse Sync Scheduler	<code>index.js → httpServer.listen</code> callback	<code>node-cron</code> (not RabbitMQ) — runs every N minutes	✓ WIRED
RabbitMQ fallback	<code>eventBus.js</code> <code>connect()</code>	Falls back to in-memory <code>EventEmitter</code> if RabbitMQ offline	✓ IMPLEMENTED

8 — Immediate Next Steps (Ordered)

1. **Fix migration 20260620300000** — Replace `dropTableIfExists('report_jobs')` in `up()` with an `alterTable` that adds `template_id` nullable and import-related columns without dropping.
2. **Commit untracked migration** — `git add`
`backend/migrations/20260619000000_nullable_template_id_report_jobs.js` and commit.
3. **Verify seed idempotency** — Ensure `backend/seeds/seed.js` uses `.onConflict('field_key').ignore()` on the `field_registry` insert so Docker re-mounts don't crash.
4. **Run migrations + seed in Docker** — After fixes above, run to activate the 24 new form fields in DB.
5. **Audit notifications.service.js** — Determine if `initSubscriptions()` duplicates `notifyHandler.init()` and consolidate.
6. **Build Bulk Import frontend page** — Template download, file upload with validation preview, confirm step.
7. **Wire filter presets to records/analytics UI** — Plug `useFilterPresets.js` hook into the `queue/analytics` filter panel.